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Attribution of extreme temperature changes during 1951–2010

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Abstract An attribution analysis of extreme temperature changes is conducted using updated observations (HadEX2) and multi-model climate simulation (CMIP5) datasets for an extended period of 1951–2010. Compared to previous HadEX/CMIP3-based results, which identified human contributions to the observed warming of extreme temperatures on global and regional scales, the current results provide better agreement with observations, particularly for the intensification of warm extremes. Removing the influence of two major modes of natural internal variability (the Arctic Oscillation and Pacific Decadal Oscillation) from observations further improves attribution results, reducing the model-observation discrepancy in cold extremes. An optimal fingerprinting technique is used to compare observed changes in annual extreme temperature indices of coldest night and day (TNn, TXn) and warmest night and day (TNx, TXx) with multi-model simulated changes that were simulated under natural-plus-anthropogenic and natural-only (NAT) forcings. Extreme indices are standardized for better intercomparisons between datasets and locations prior to analysis and averaged over spatial domains from global to continental regions following a previous study. Results confirm previous HadEX/CMIP3-based results in which anthropogenic (ANT) signals are robustly detected in the increase in global mean and northern continental regional means of the four indices of extreme temperatures. The detected ANT signals are also clearly separable from the response to NAT forcing, and results are generally insensitive to the use of different model samples as well as different data availability.

Keywords Detection and attribution · Extreme temperature · Anthropogenic forcing · Natural variability · CMIP5 models

1 Introduction

Weather and climate extreme events exert devastating impacts on human society, the economy, and ecosystems, so better understanding of observed changes in climate extremes is fundamental for more reliable future prediction of their changes and associated impacts. Extreme temperatures that are associated with heat waves and cold snaps have warmed globally over the last several decades (Alexander et al. 2006; Donat et al. 2013; Hartmann et al. 2013). The observed increases in hot extremes and decreases in cold extremes are, by and large, in line with the observed shift of daily temperature distributions under global warming (Donat and Alexander 2012). The IPCC Special Report on Extremes (SREX) indicated that it was *likely* (66–100 % probability) that anthropogenic influences had contributed to the observed changes at the global scale (Seneviratne et al. 2012) but the SREX did not attribute causes to observed chances on regional scales. Several recent formal

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Table 1 List of CMIP5 model simulations analyzed in this study

Model	ALL	NAT	CTL (No. of 50-year chunks)
bcc-csm1-1	1	1	10
CanESM2	5	5	20
CCSM4	3	–	2
CNRM-CM5	1	6	–
CSIRO-Mk3-6-0	10	5	10
GFDL-ESM2M	1	–	10
HadGEM2-ES	4	4	10
IPSL-CM5A-LR	4	3	20
IPSL-CM5A-MR	–	3	–
MIROC-ESM	1	–	12
MIROC-ESM-CHEM	1	–	4
MRI-CGCM3	1	–	10
NorESM1-M	1	1	10
Total	12 models, 33 runs	8 models, 28 runs	11 models, 118 chunks

The numbers of ensemble members are given for ALL and NAT runs (the same number of runs were used for each of the three 50-year periods: 1951–2000, 1956–2005, and 1961–2010). For CTL (preindustrial control) runs, the number of 50-year chunks are provided

detection and attribution studies have further identified human and/or natural influences on the observed changes in extreme temperatures (Christidis et al. 2011, 2014; Zwiers et al. 2011; Min et al. 2013; Morak et al. 2013; Sun et al. 2014). Utilizing rigorous statistical techniques (called optimal fingerprinting; see below), anthropogenic signals (ANT, mainly due to increases in greenhouse gases) have been detected in observed changes in the temperature of both the warmest and coldest extremes on a global scale and also on continental and regional scales in some regions since the mid-twentieth century. These new studies have provided a strengthening of the evidence for human influence on temperature extremes, which the IPCC AR5 report assessed as *very likely* (90–100 % probability; Bindoff et al. 2013).

Although several attribution studies are available, further analyses are needed to assess robustness of the conclusions to possible uncertainty factors. In particular, the separation of human influence from natural influences on observed changes remains limited because of uncertainties related to inter-model differences. Previous regional detection studies (e.g., Min et al. 2013) were based on rather limited number of models from the Climate Model Intercomparison Project Phase 3 (CMIP3, Meehl et al. 2007) archive that provided the necessary daily output that is required to study changes in extremes. More importantly, observations and multi-model simulations have been updated, providing datasets up to 2010 with better observed spatial coverage.

Given the limitation of previous studies and newly available datasets, this study aims at performing updated attribution analyses of temperature extremes. We compare changes in extreme temperature from updated observations to those in recent model experiments that provide both natural forcing-only runs and natural-plus-anthropogenic forcing runs. In order to examine influence of the use of different datasets and/or different periods on detection results, sensitivity tests are conducted, from which individual impacts are isolated.

The structure of this paper is as follows. Section 2 describes datasets from observations and model simulations and the analysis methods including processing of extreme indices and an optimal fingerprinting technique. In Sect. 3, after making qualitative comparisons of observed and simulated long-term variations in extreme temperatures, detection and attribution results are provided for various spatial domains from global to continental domains. The paper concludes in Sect. 4 with a summary and discussions.

2 Data and methods

We have obtained extreme temperature indices (annual minima/maxima of daily minimum/maximum temperatures: TNn, TNx, TXn, TXx) from the HadEX2 observational dataset (Donat et al. 2013) and Coupled Model Intercomparison Project Phase 5 (CMIP5, Taylor et al. 2012) multi-model datasets from historical (ALL), historicalNat (NAT), and preindustrial control (CTL) experiments (Table 1). Observations and ALL and NAT simulations provide datasets for 1951–2010, which are divided into three 50-year periods for analysis: 1951–2000 (referred to as 5100), 1956–2005 (5605) and 1961–2010 (6110). Not all CMIP5 ALL simulations have been extended to 2010 or later, and thus where necessary, we extend historical ALL runs (1951–2005) with the corresponding representative concentration pathways (RCP) 4.5 runs (2006–2010) while for NAT we only use models that provide historical NAT runs for an extended period up to 2010. Results from the 5100 period can be readily compared with those from Min et al. (2013) which were based on the older versions of observations (HadEX, Alexander et al. 2006) and CMIP3 multi-model datasets. This comparison based on this period (5100) will be useful to understand the influences of different datasets. Results from 5605 and 6110 will give insights into influence of more recent changes in both observations and models on detection results.

Min et al. (2013) used ALL and anthropogenic-only (ANT) simulations that were composed of available samples of models; that is, the ALL simulations came from a collection of models that was not identical to the collection of models that produced the ANT simulations used in that study. In this study, we use 13 models from the CMIP5 archive, 7 models of which provide both ALL and NAT runs (Table 1). Restricting the analysis to the latter multi-model selection will remove possible confounding influences that could arise due to the use of different models when estimating response patterns to the external forcing (ALL or NAT here). To check possible influence of this model selection, we first use the full set of models that ran each forcing (12 models, 33 runs for ALL, and 8 models, 28 runs for NAT) and then test the sensitivity of attribution results to the use of only the 7 models that ran both types of simulations. Ensemble means are first calculated for individual models, and then multi-model means are obtained by taking arithmetic averages of the model-specific ensemble means, which gives equal weight to each model when constructing multi-model means. Results were essentially the same when we weighted runs equally (as opposed to models) to calculate multi-model means. ANT responses are estimated from ALL minus NAT, which assumes that the responses to historical ANT and NAT forcing are additive. We use CTL simulations from the 11 models (two models do not provide CTL simulation data) to estimate the characteristics of internal climate variability, which provide 118 non-overlapping 50-year chunks in total (Table 1, also see below).

Analysis methods are identical to those used in Min et al. (2013). We first standardize extreme temperature indices (TNn, TXn, TNx, and TXx) by converting them onto probability-based indices (PIs) ranging from 0 to 1 based on the Generalized Extreme Value (GEV) distribution (Min et al. 2011; Zhang et al. 2013). Note that other standardization methods, such as replacing temperature extremes with ranks, would be equally effective (Zhang et al. 2013). This procedure allows inter-comparison of extremes whereby equal weight is given to extremes at all locations; it is carried out on the original grids of the observations and individual models. The resulting standardized PIs are then interpolated onto the HadEX2 grid (3.75° longitude by 2.5° latitude) and masked with the observed space–time data availability for each of the 50-year periods (5100, 5605, and 6110). After calculating PI anomalies on each grid by removing the time average at each location, we take area averages of PI anomalies over the globe (GLB) and five continental domains of North America (NAM), South America (SAM), Europe (EUR), Asia (ASI), and Australia (AUS) where spatial data coverage is more than 50 %. Finally, in order to reduce noise on inter-annual time scales we obtain 5-year mean time series of area-averaged PI anomalies, which results in 10-dimensional data vectors of observations and model simulations (ALL, ANT, NAT, and CTL).

In order to compare observed and modeled PI time series, we employ an optimal fingerprinting technique (Allen and Stott 2003) where observations ($\mathbf{y}$) are regressed onto multi-model mean response patterns ($\mathbf{X}$, fingerprints of ALL, ANT and NAT here) such that $\mathbf{y} = (\mathbf{X} - \boldsymbol{\nu})\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Here regression coefficients $\boldsymbol{\beta}$ (or scaling factors) are obtained by the total least squares method, $\boldsymbol{\varepsilon}$ represents residual variability that is generated internally in the climate system, and $\boldsymbol{\nu}$ represents the component of $\mathbf{X}$ due to internal variability that remains after multi-model averaging. The variance–covariance matrix of $\boldsymbol{\varepsilon}$ is estimated from CTL runs, and that of $\boldsymbol{\nu}$ is taken to be proportional to the variance–covariance matrix of $\boldsymbol{\varepsilon}$, where the constant of proportionality reflects the methods used to calculate the multi-model ensemble response patterns. There are two regression settings. Single-signal analyses are conducted by regressing observations onto single (ALL, ANT or NAT) fingerprints so as to examine whether the signal considered is present in the observed changes. In the two-signal analysis, observations are regressed onto ANT and NAT simultaneously, from which one can examine whether ANT and NAT are jointly detected and whether the influence of ANT is separable from that of NAT and internal variability in the observations. The 50-year chunks of CTL simulations are divided into two sets (59 chunks each; see Table 1) for this analysis. One set is used to obtain best estimates of $\boldsymbol{\beta}$ and the other set is used to estimate the 5–95 % uncertainty range for $\boldsymbol{\beta}$ and also to carry out a standard residual consistency test (Allen and Tett 1999; Allen and Stott 2003). This test provides a useful guide to check whether model-simulated internal variance is consistent with observational residual variance. Since data vectors have low dimension compared to the number of chunks of CTL simulations available for covariance matrix estimation (10 dimensions vs. 59 chunks for each of the two covariance matrix estimates), this study does not apply further dimension reduction such as empirical orthogonal function (EOF)

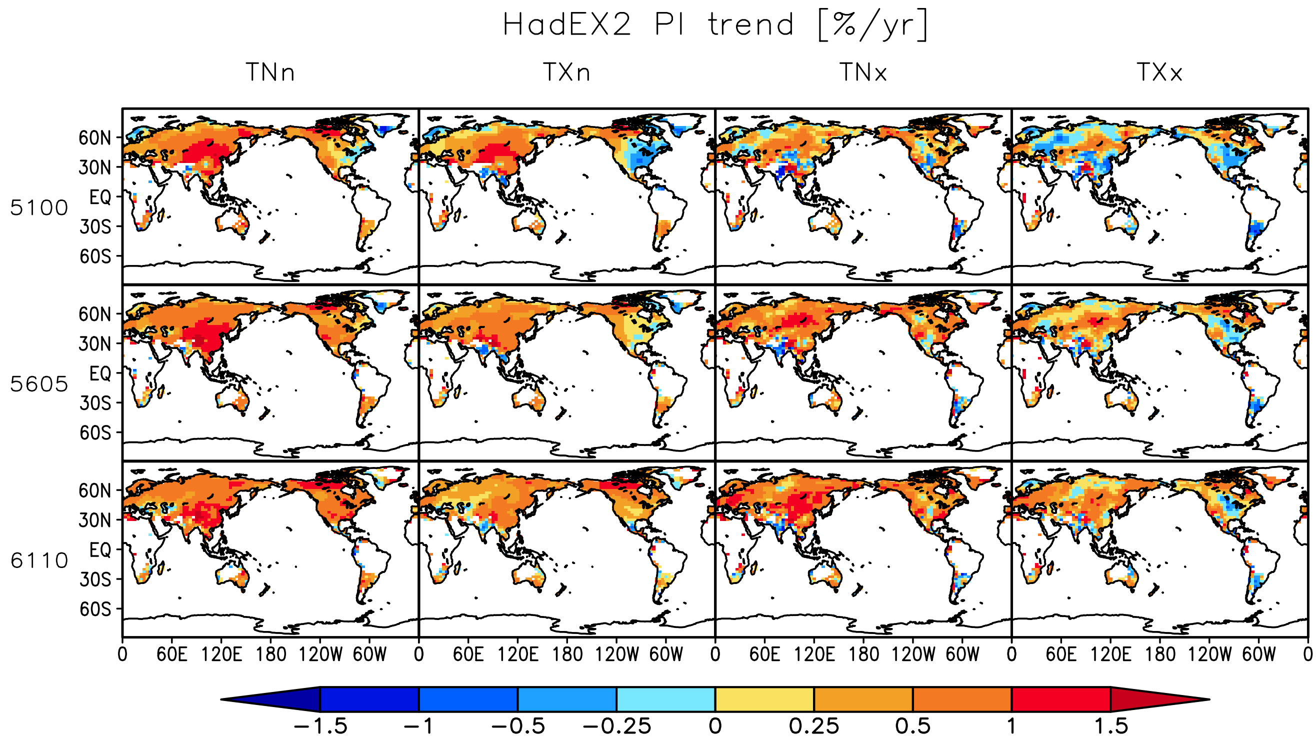

Fig. 1 Geographical patterns of observed trends in PI of temperature extremes (% probability per year) during three 50-year periods (1951–2000; 5100, 1956–2005; 5605, and 1961–2010; 6110). *Row* and *column* indicate three periods and four extreme temperature indices (TNn, TXn, TNx, and TXx), respectively. Positive trends mean weakening of cold extremes in TNn and TXn and intensification of warm extremes in TNx and TXx. Analysis is confined to grid points with 40 years or more of data for each variable for each period. The observed TNx mask has been applied to all variables for consistency

truncations (Allen and Tett 1999). The confidence intervals for the scaling factors are obtained following Ribes et al. (2013).

3 Results

3.1 Observed and modeled trends

Figure 1 illustrates observed patterns of PI trends for the four extreme indices over three periods. Positive trends mean weaker cold extremes and stronger warm extremes. The magnitude of the trends is substantial over broad areas, and locally often exceeds 0.5 % (PI trends expressed in % probability) per year, or 25 % over the 50-year periods of record, indicating a strong tendency for the warmest warm and cold extremes to occur near the end of the record. Overall positive trends are dominant in cold extremes (TNn and TXn) during the three periods, although there are large areas of decreasing trends in TXn in the 5100 period, in particular over eastern North America where average temperatures also show a so-called ‘warming hole’ (Portmann et al. 2009). Over the three periods, warming trends of cold extremes in the Asia region tend to weaken, while they strengthen over North America. For the warmest days and nights (TNx and TXx), a mixture of positive and negative trends is apparent during 5100, but positive trends occur over larger regions during the 5605 and 6110 periods. The 5100 trend patterns are consistent with those in Min et al. (2013) where HadEX data were used, suggesting weak influence of the use of different datasets in regions where both HadEX and HadEX2 provide data coverage.

Trend patterns in PI simulated by the ALL and NAT multi-models are displayed in Figs. 2 and 3, respectively. Trends are first calculated for individual models and then averaged to estimate multi-model mean trends. These maps are spatially smoother than the corresponding maps for the observations (Fig. 1) because multi-model averaging has removed a substantial amount of the influence of internal climate variability. The historical ALL runs exhibit warming almost everywhere for all extreme indices and all three periods, meaning weakening of cold extremes and intensification of warm extremes responding to the combined natural-plus-anthropogenic forcings. Trends in cold extremes tend to increase more over high latitudes while warm extremes exhibit stronger trends over the subtropics. This difference in trend patterns between cold and warm extremes is generally consistent with CMIP3 results (Min

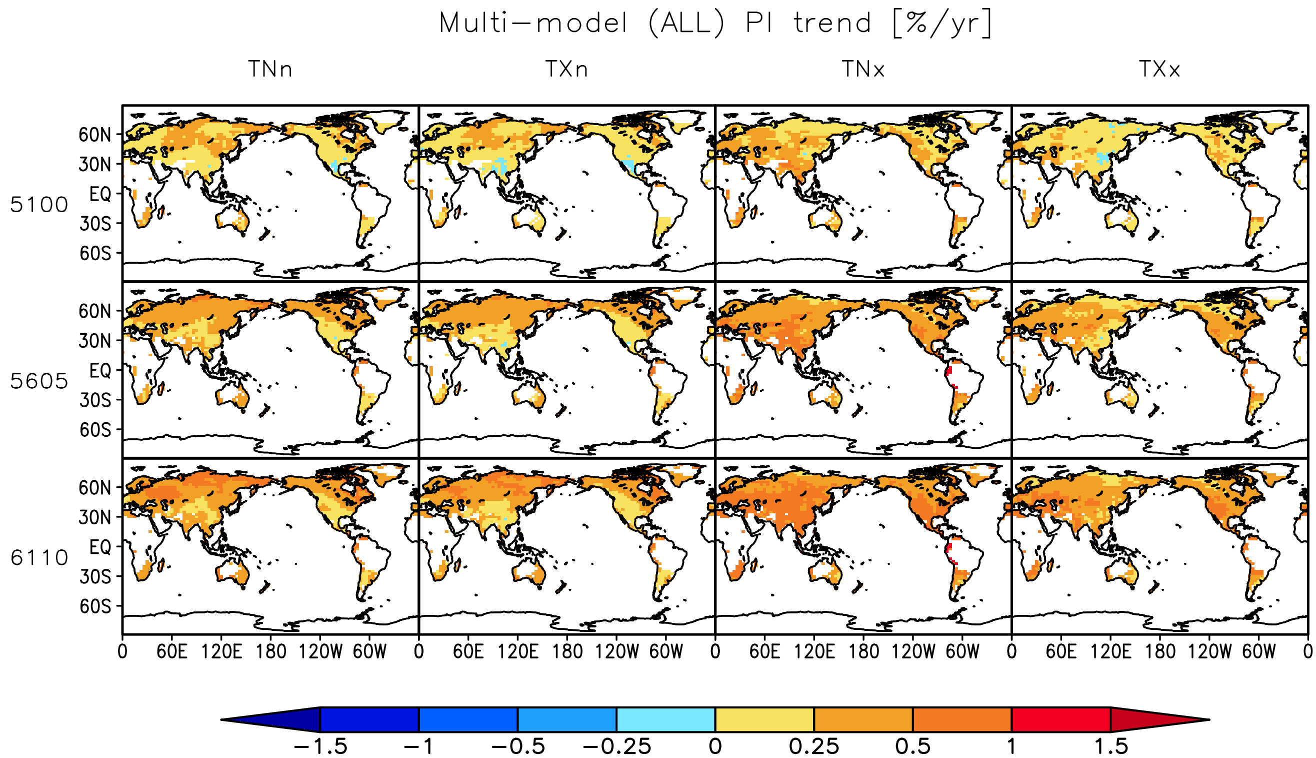

Fig. 2 As in Fig. 1, but for multi-model mean PI trends (% probability per year) from ALL simulation. Refer to Table 1 for the list of model runs

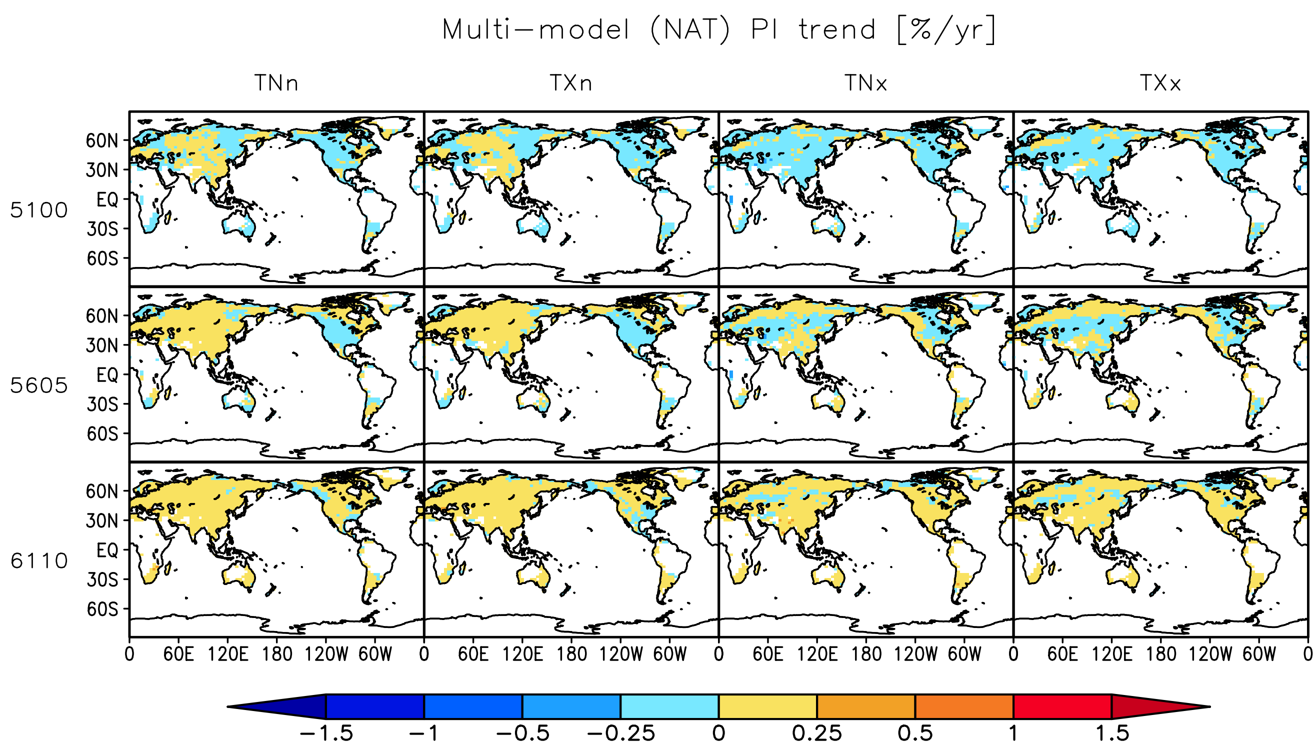

Fig. 3 As in Fig. 1, but for multi-model mean PI trends (% probability per year) from NAT simulation. Refer to Table 1 for the list of model runs

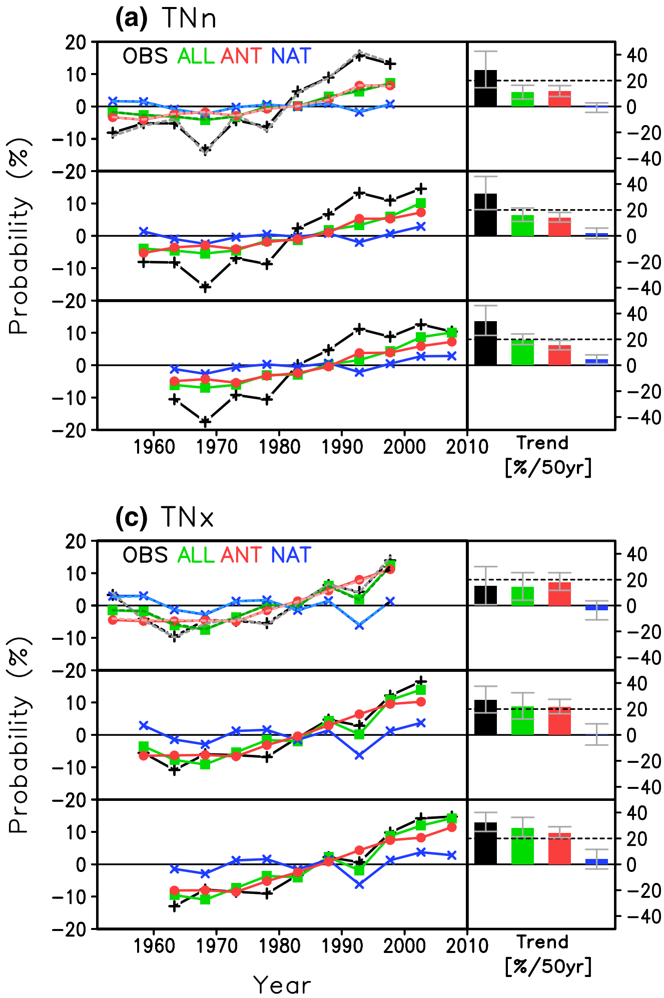

Fig. 4 Global-mean 5-year mean PI (unit: % probability) anomaly time series (*left*: *line plot*) and trends (*right*: *color bar plot*) for three periods for TNn, TXn, TNx and TXx from HadEX2 observations (OBS, *black*) and CMIP5 multi-model means of ALL (*green*), ANT (estimated from ALL-NAT, *red*), and NAT (*blue*) simulations. *Grey error bars* indicate 95 % confidence intervals of linear trends. *Grey*, *pink*, *dark green* and *sky-blue dashed lines* for the 5100 period indicate results when HadEX data coverage is applied to observations and CMIP5 model runs

et al. 2013). It is clearly seen that warming trends become stronger for more recent periods, with large areas again exhibiting increases in PI exceeding 25 % over the period of record, as in observations (Fig. 1). NAT runs exhibit large areas with negative trends for the four extreme indices in the 5100 period (Fig. 3), consistent with the slight net cooling effect due to volcanic forcing for the period (Huntingford et al. 2006; Min et al. 2013). The positive areas gradually increase in the 5605 period and then weak warming expands over the entire domain in the 6110 period. This warming trend in NAT runs seem to be due to the combined influences of solar and volcanic activity (Jones et al. 2013).

Global-mean 5-year mean anomaly time series of extreme temperature indices (standardized PI in % probability) for the three periods are shown in Fig. 4. Anomalies are relative to each time mean. Observed (HadEX2) and simulated (CMIP5 ALL, ANT and NAT multi-model means) changes are compared for TNn, TXn, TNx and TXx. For the 5100 period, observed changes are generally characterized by a slight cooling in the early decades and a strong warming after the 1970s with the biggest increase occurring in TNn (more than 20 % over the 50 years). ALL results also show warming trends although the simulated magnitude of change is weaker than observed in cold extremes (TNn and TXn), particularly for TNn, and warming rates are less variable in time. ANT runs exhibit similar changes to ALL runs. NAT simulations display small amplitude decadal scale variations that may reflect the effects of volcanic forcing (e.g., the period 1991–1995 appears to be somewhat cooler for all indices). Overall, these observed and simulated changes are consistent with those reported in Min et al. (2013) based on HadEX data and CMIP3 simulations.

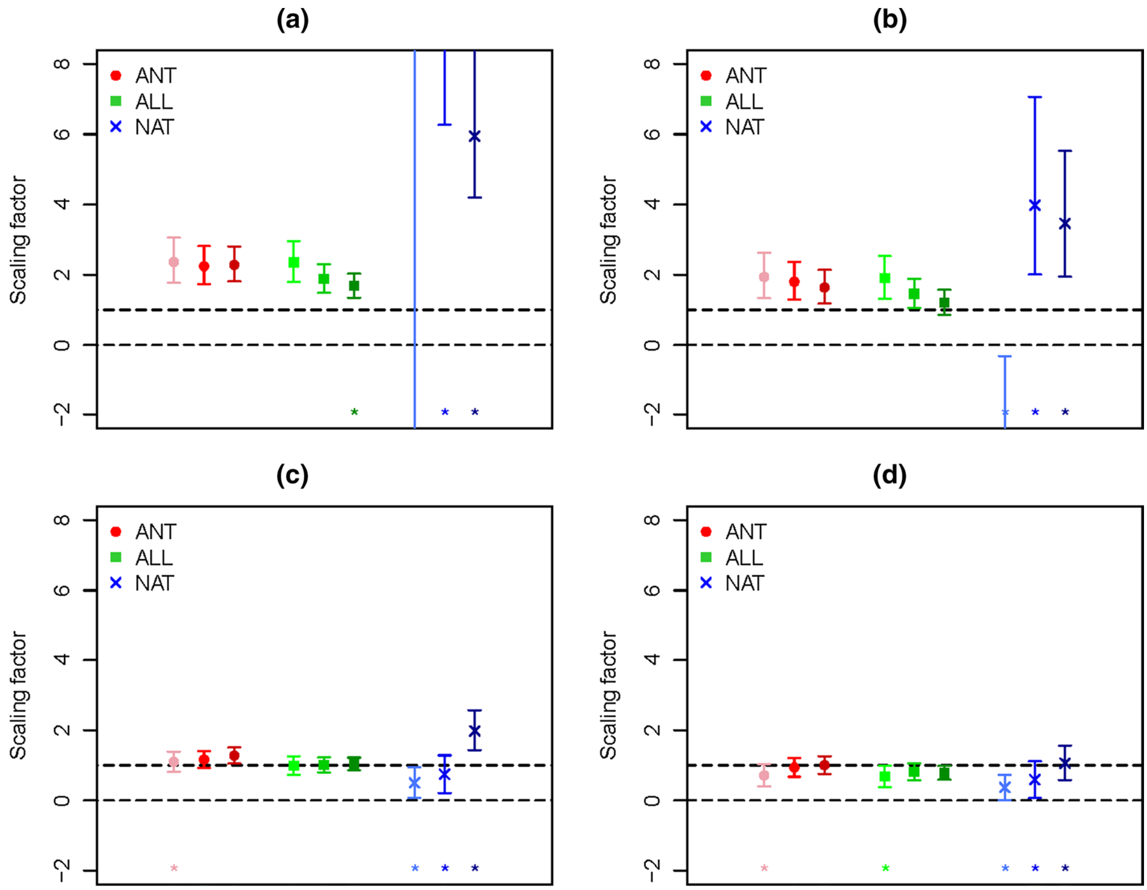

Fig. 5 Results from single-signal optimal detection analyses of global mean PI time series for **a** TNn, **b** TXn, **c** TNx and **d** TXx. Best estimates (*marks*) and 5–95 % ranges (*error bars*) of scaling factors are shown for three 50-year periods of 1951–2000 (*light*), 1956–2005 (*medium*), and 1961–2010 (*dark color*), respectively. An *asterisk* indicates the failure of the residual consistency test because of too small model variability

Interestingly, there is a notable difference in warm extremes (TNx and TXx) from ALL runs between CMIP5 and CMIP3 results for the ensembles of runs we used in this study and in Min et al. (2013). While CMIP3 models with natural and anthropogenic forcings used by Min et al. (2013) underestimate the temperature increase in warm extremes (Fig. 4 of Min et al. 2013), the CMIP5 runs used here display larger increases, capturing observed changes with similar amplitude (Fig. 4). Some factors that might account for this difference include different responses to NAT forcing, i.e., stronger cooling of warm extremes in CMIP3 than in CMIP5, and differences or improvements in land surface schemes and other associated physics such as soil moisture and precipitation (cf. Mueller and Seneviratne 2011), which warrants further investigation. The differences reported here are calculated from model output that has been masked according to the observational availability, and thus different data coverage between HadEX2 and HadEX datasets (larger coverage in HadEX2, Donat et al. 2013) could also be a factor. To evaluate this possibility, we apply HadEX data coverage to HadEX2 and all CMIP5 data and recalculate global mean changes (underlying pale dashed lines in the upper panel of each part of Fig. 4). It is clearly seen that there is almost no difference between solid lines and dashed lines for all observed and simulated extreme indices, confirming negligible influence of different data coverage for the latter half of twentieth century (Donat et al. 2013).

The main update of this study comes from adding the recent 10 years (2001–2010) and testing the sensitivity of detection results. For this purpose we shift the 50-year analysis window by 5 years, i.e., 1956–2005 and 1961–2010, from each of which 5-year means are obtained.

Moving the analysis window ahead 5 years produced differences in the 5605 period relative to the 5100 series that can be seen in observations as well as in model results, depending on the specific extreme indices considered. Firstly, observed warming trends in both cold extremes (TNn and TXn) and warm extremes (TNx and TXx) are strengthened with stronger amplitudes in warm extremes,

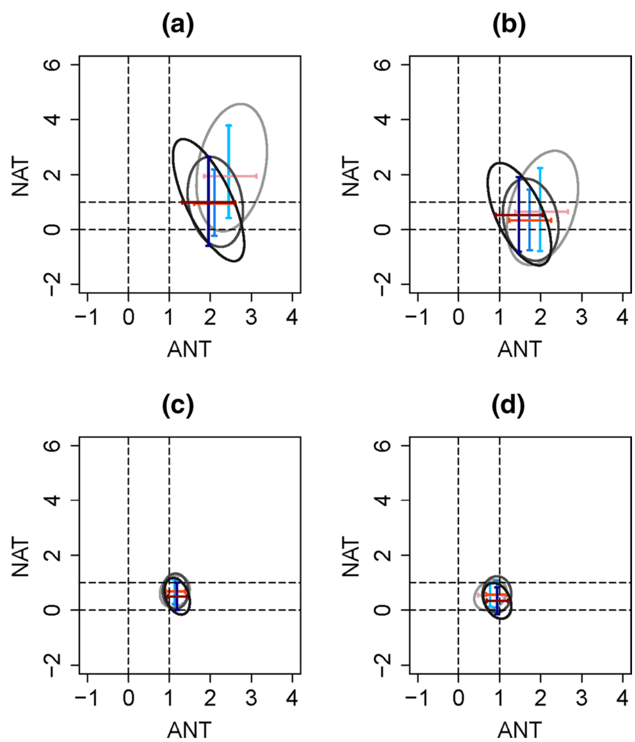

Fig. 6 The joint 90 % uncertainty range for the ANT and NAT scaling factors when temperature extreme indices are regressed onto ANT (*x* axis) and NAT signals (*y* axis) simultaneously: **a** TNn, **b** TXn, **c** TNx and **d** TXx. The *error bars* indicate the marginal 5–95 % ranges of the scaling factors for each forcing. *Light*, *medium* and *dark colors* indicate 1951–2000, 1956–2005, and 1961–2010 periods, respectively. The *dashed horizontal/vertical lines* represent zero and unity

consistent with trend patterns (Fig. 1). For 5605, the largest long-term change in observations occurs in TNx and the increase in TXx also becomes clearly apparent, showing nearly a 10 % increase during the last decades, which is comparable to the increase in TXn. This is in a good agreement with recent literature (Seneviratne et al. 2014; Sillmann et al. 2014) that demonstrates a continued increase of hot extremes over land despite the recent pause in global mean temperature (the so-called global warming 'hiatus'). Secondly, in contrast to observed asymmetric changes between cold and warm extremes, model (ensemble average) results show continued and stronger warming in both cold and warm extremes during the recent two decades. There are bigger increases in warm extremes than in cold extremes, well reproducing observed changes. Thirdly, the NAT effect of long-term cooling disappears during 1956–2005 and instead it brings decadal fluctuations with relatively cooler conditions centered around 1970 and 1990 and warmer conditions around 1960 and 2000. Consequently, ANT series (estimated from ALL–NAT) display very similar trends to ALL runs in warm extreme indices, differently from the 1951–2000 results (Min et al. 2013).

In the 6110 period, a pause in the observed warming trends of the cold extremes becomes apparent, but is presumably limited to a band in the northern mid-latitudes as shown by Sillmann et al. (2014). The simulated long-term change seen in ALL runs is still weaker than observations in cold extremes while the observed increase in warm extremes is well reproduced by the models, especially in TXx. NAT simulations display slight warming trends in the 6110 period due, likely due to recovery from recent volcanic eruptions.

3.2 Detection results

Qualitative comparisons of global-mean time series made above suggest that when looking at more recent trends or including changes in recent years we may expect better agreement between observed and simulated changes in temperature extremes, particularly for warm extremes. We quantify this comparison and also try to isolate the human contribution to the observed changes in extreme temperatures.

Global and continental mean PI time series from observations and simulations (ALL, ANT and NAT) for three periods are compared using the optimal fingerprinting technique described previously. Figure 5 shows results from single-signal detection analyses of global mean PI series where observations are regressed onto each of the external forcings (ALL, ANT and NAT). Best estimates and 5–95 % ranges of regression coefficients $\boldsymbol{\beta}$ (or scaling factors) are presented for three periods. Signals of external forcings are assessed to be present in the observations if the 90 % uncertainty ranges of the scaling factors lie above zero (detection). Additionally, if the 90 % ranges for $\boldsymbol{\beta}$ of the detected signals include unity, the observed changes are assessed to be consistent with the fingerprints of external forcing (attribution) provided that other influences on the observations cannot provide a plausible explanation for the observed changes. The results show that both ALL and ANT signals are robustly detected for the four extreme indices over all periods, with the residual consistency test passed in almost all cases of ALL and ANT results (21 out of 24 cases). The influence of NAT is sporadically detected, but scaling factors vary greatly across extremes indices and analysis periods, and thus NAT results cannot be considered to be robust. Best estimates of scaling factors for ANT signal are near 2 for TNn, implying that models underestimate the observed warming (weakening) of the coldest night annually by a factor of 2 at the global scale over the three periods. ANT and ALL results are similar for 5100, while the scaling factors for the ALL signal are closer to unity for 6110, indicating better agreement with observations. Detection results for TXn and TNx are similar, but with best estimates being closer to unity; scaling factors

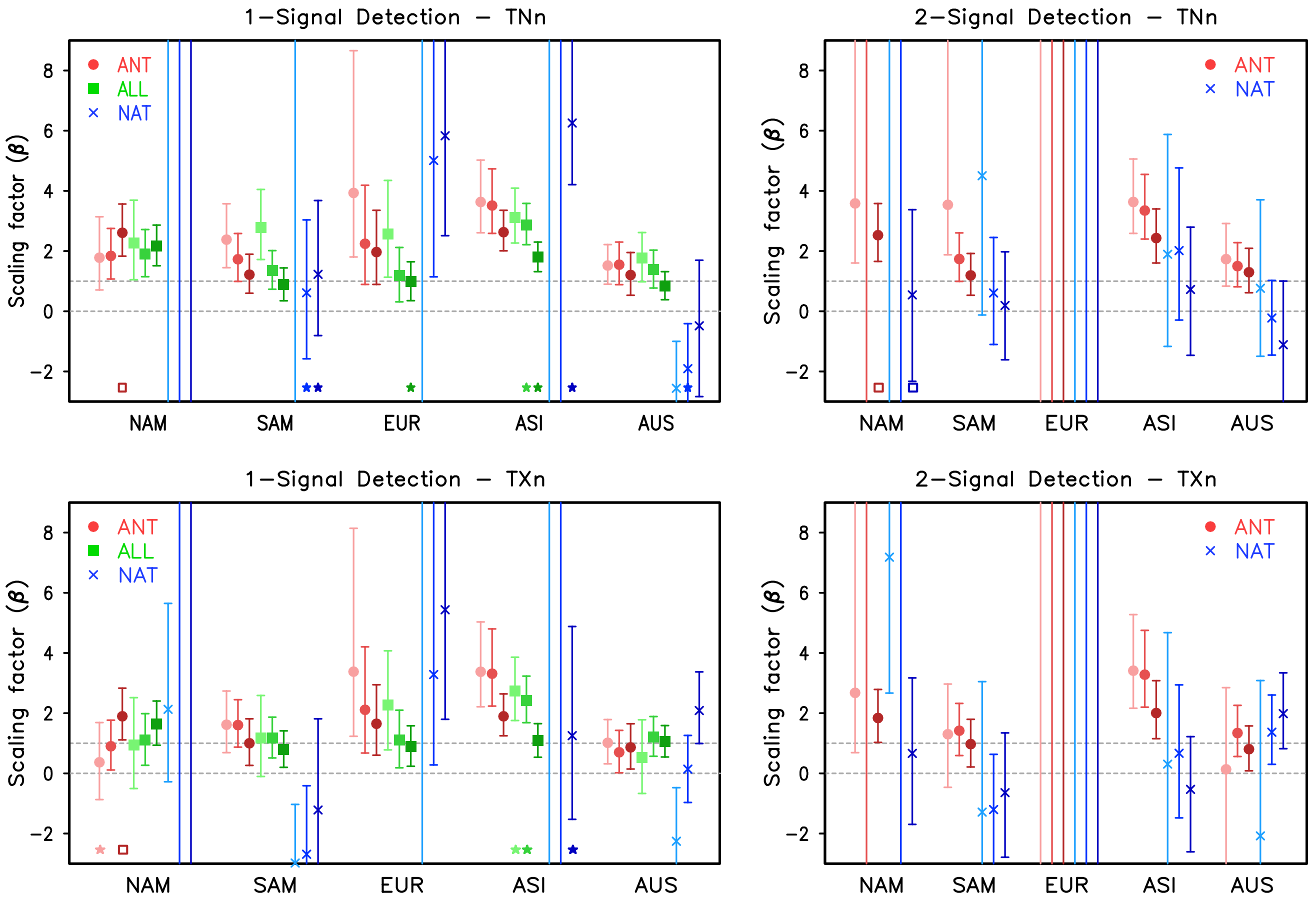

Fig. 7 Result from (*left*) single-signal and (*right*) two-signal detection analyses for cold extreme indices (*top*) TNn and (*bottom*) TXn for continental scales. Results for individual indices are based on time vectors of area-mean 5 years-mean PI. Best estimates (*marks*) and 5–95 % ranges (*error bars*) of scaling factors are shown for three periods: *light*, *middle* and *dark colors* indicate 1951–2000, 1956–2005, and 1961–2010 periods, respectively. An *asterisk* (*square*) indicates the failure of the residual consistency test due to too small (too large) model variability

are very close to unity for TNx, consistent with time series of global mean PIs in Fig. 4c. ANT and ALL are detected in TXx, albeit with some indication that models slightly overestimate observed changes in TXx over the three periods. These results are consistent with those of Min et al. (2013) for 5100, who also found model underestimation of changes in TNn and TXn, good agreement between observations and ALL simulations in TNx, and model overestimation of changes in TXx. Zwiers et al. (2011), using a different detection and attribution technique, a shorter period (1961–2000), and 7 CMIP3 models, also obtained broadly consistent detection results for all four extreme temperature indices.

In order to consider the combined influence of ANT and NAT simultaneously, a two-signal analysis has also been carried out. Figure 6 shows the resulting joint 90 % uncertainty ranges for the ANT and NAT scaling factors. The 90 % uncertainty contours exclude the origin (0, 0) for all temperature extremes and all periods, meaning that ANT and NAT are jointly detected through two-way regression. However, for cold extremes (TNn and TXn), only ANT is consistently detected based on the marginal 90 % uncertainty ranges of the scaling factors with best estimates of the ANT scaling factor larger than unity, indicating model underestimation of changes in cold extremes by a factor of 2 or so. In contrast, both ANT and NAT are detected in warm extremes in 5100 and 5605. Also, ANT scaling factors for warm extremes are close to unity, implying better agreement with observations in warm seasons, although models appear to overestimate the NAT response (i.e., too strong volcanic cooling in summer).

The results from detection analyses for cold extremes averaged over five continental areas of NAM, SAM, EUR, ASI, and AUS for the three periods are plotted in Fig. 7. Single-signal analyses of TNn indicate that ALL and ANT signals are robustly detected in all continental domains and all periods while NAT is not detected except for EUR (5605 and 6110) and ASI (6110). In the 5100 period, best

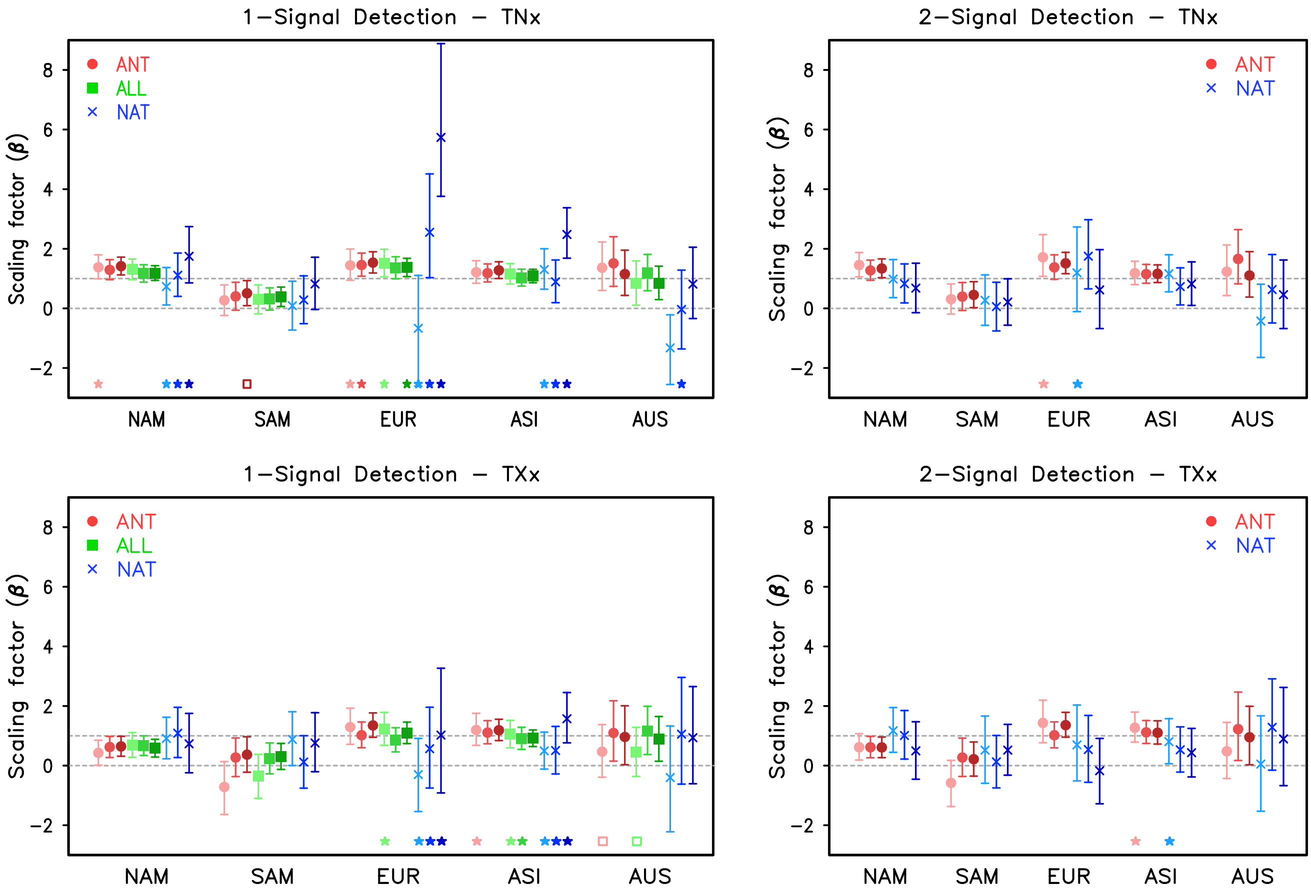

Fig. 8 As in Fig. 7, but for warm extreme indices (*top*) TNx and (*bottom*) TXx

estimates of detected ALL and ANT signals are larger than unity over all continents whereas they are closer to unity for the 6110 period, except for NAM. These features are also found in TXn results, including better agreement between ALL and observations in 6110, although ANT is not detectable in NAM for 5100. Two signal analyses (Fig. 7, right panels) indicate that the detected ANT signals are separable from NAT over four continents (NAM, SAM, ASI and AUS) for TNn and TXn over all periods, except for NAM (5605 in TNn and TXn), SAM (5100 in TXn) and AUS (5100 in TXn). Best estimates of the detected ANT signal scaling factors are larger than unity over NAM and ASI, indicating model underestimation of the anthropogenic warming of cold extremes over these continents. ANT scaling factors approach unity more closely in more recent periods, especially over ASI. Continental detection results for warm extremes are displayed in Fig. 8. ALL signals are detected robustly over four continental domains (NAM, EUR, ASI, AUS) for both indices (TNx and TXx) in the single-signal analyses, and ANT signals are clearly detected and separable from NAT over the same domains in the two-signal analyses over all periods, except for AUS (5100). Also, best estimates of the detected ALL and ANT signal scaling factors are very close to unity, particularly for TNx. Models overestimate the ALL response in TXx over NAM and SAM. Overall, the continental-scale detection results for warm extremes are consistent over different periods as in the corresponding global cases (Figs. 5, 6).

We also tested the sensitivity to the use of the 60-year period 1951–2010. Detection results are found to be very similar to those from the 6110 period for both global and continental-scale extreme temperatures (not shown). This suggests an important contribution of the observed changes during the recent decade (2001–2010) to the long-term (50-year or longer) trends, as reflected by better detection for 6110. In addition it means that observed long-term trends are less affected by internal climate variability when including the recent decade (see below).

3.3 Sensitivity tests

The results shown above (Figs. 5, 6, 7, 8) are based on ALL and NAT signals estimated from all available models and thus the estimated ANT signal may be confounded with the influence of model differences (see Table 1). We therefore also evaluate the robustness of our detection results

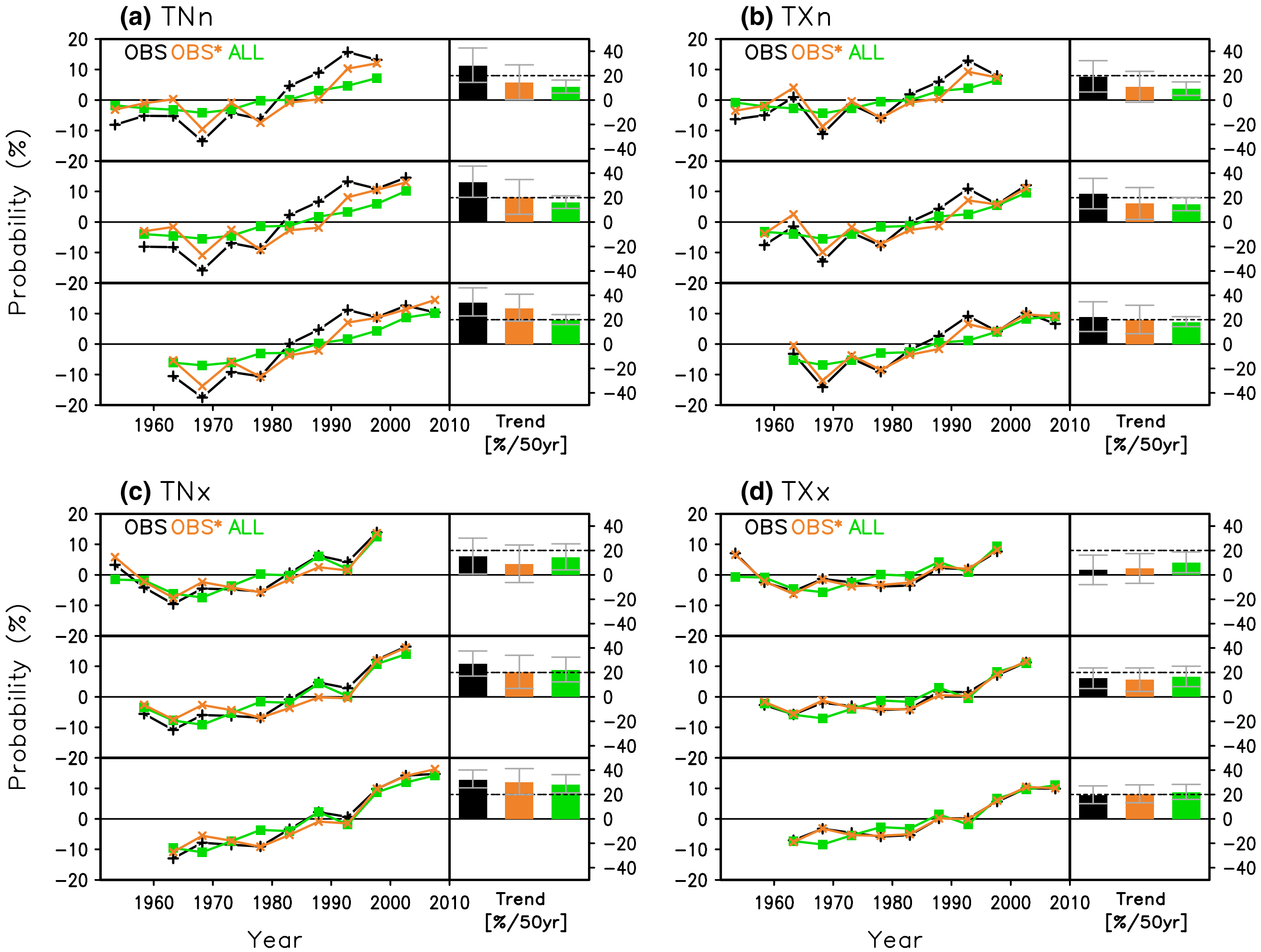

Fig. 9 As in Fig. 4, but showing time series of AO and PDO residual observations (OBS*, *orange color*) in comparison with ALL simulations

by redoing our analysis using only the seven models that provided both ALL and NAT runs (bcc-esm1-1, CanESM2, CNRM-CM5, CSIRO-Mk3-6-0, HadGEM2-ES, IPSL-CM5A-LR, NorESM1-M; Table 1) to estimate ALL and NAT signals. Detection results using signals from this subset of models are found to be essentially same as those from the full model case for both global and continental extreme temperatures (not shown), suggesting insensitivity of our findings to the inter-model differences.

Recent detection and attribution studies (Zwiers et al. 2011; Christidis et al. 2011; Min et al. 2013; Morak et al. 2013) consistently show that long-term warming of cold extremes is underestimated by the models on global to continental scales, as seen above. On the other hand, it has been noted that, in contrast to observed changes in cold extremes over the Northern mid-latitudes during the recent 15 years, models warmed cold extremes in these regions (Sillmann et al. 2014), suggesting a possible influence of the recent hiatus in the warming of global mean temperature on regional temperature extremes. Various studies have linked the warming hiatus to the state of the eastern Pacific Ocean, for example, via the recent negative phase of the Pacific Decadal Oscillation (PDO, Trenberth and Fasullo 2013) or the Interdecadal Pacific Oscillation (IPO, Meehl et al. 2014), which strongly affects Northern mid-latitude temperature during winter (Trenberth et al. 2014). The observed trends in cold extremes may also be affected by the Arctic Oscillation (AO, Brown et al. 2008), the phase of which exhibits similar variations to the observed variation in cold extremes. Therefore, we removed PDO and AO influences from the four observed temperature extreme indices and then repeated the optimal fingerprinting analyses to test the sensitivity of detection results to possible influence of natural variability. In order to keep all calculations within the scope of extreme value theory, we have removed the PDO and AO influence from the observed PI time series by including PDO and AO indices as covariates in the GEV model that is used to obtain PI. To this end, we

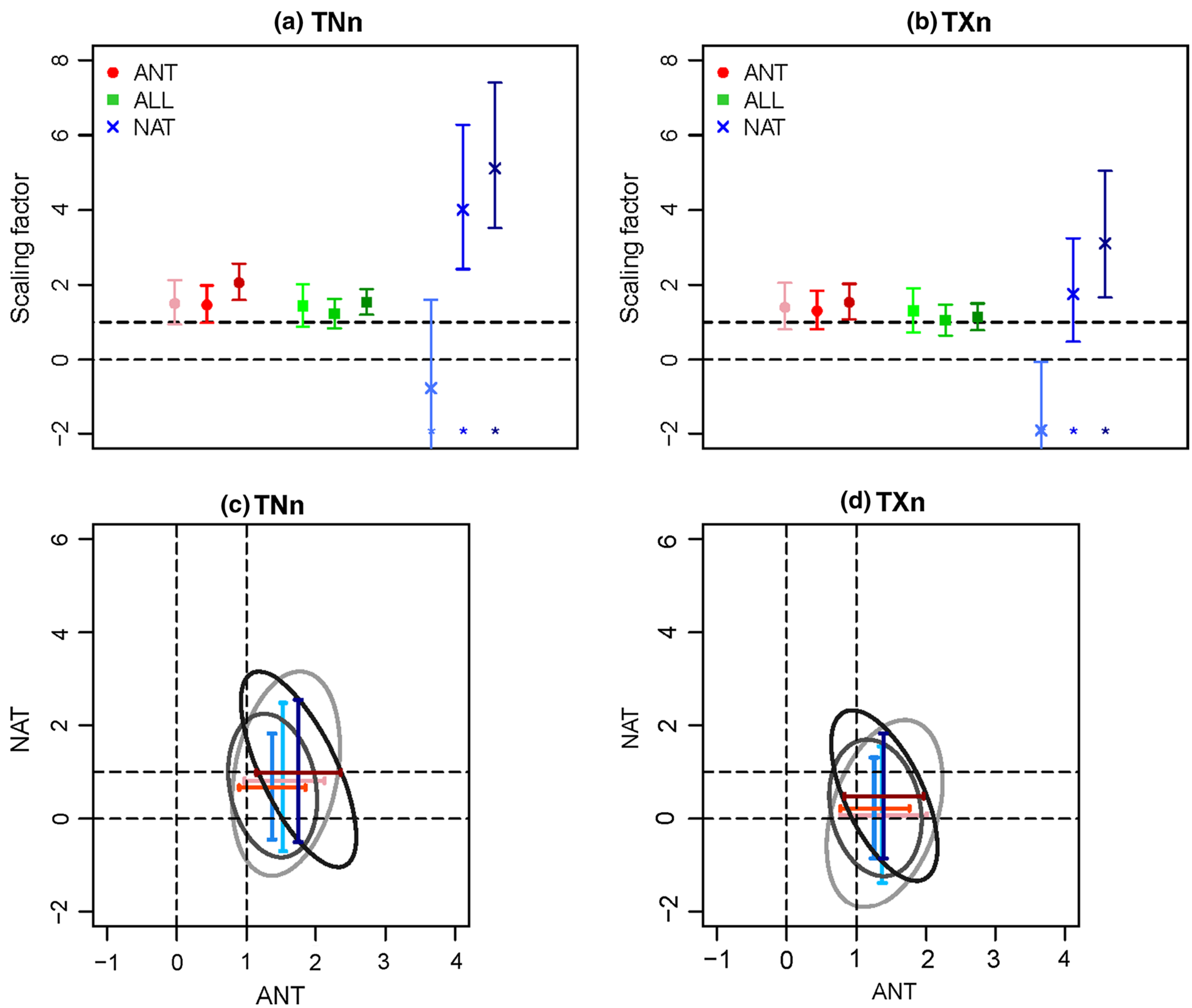

Fig. 10 As in Figs. 5 and 6, but using AO and PDO residual observations: (*top*) single-signal and (*bottom*) two-signal analyses for cold extreme indices of (*left*) TNn and (*right*) TXn

made the location and scale parameters of the GEV distribution dependent upon the PDO and AO index time series, and then calculated annual PI series by using the PDO- and AO-adjusted GEV parameters (Min et al. 2011).

Figure 9 shows time series of the extreme temperature indices before and after removing AO and PDO influences on the observations. Removal of these influences warms TNn in the early decades (1951–1980) relative to the unadjusted observations and results in more consistent warming subsequently, much reducing the hiatus-like behavior in recent decades. Similar, albeit weaker, features can be seen in TXn and TNx during 5100 and 5605, while TXx is not significantly affected by the AO and PDO. The detection results reflect these changes in the observations (Fig. 10). For cold extremes, best estimates of the detected ALL and ANT signals are very close to unity, particularly for the 5100 period at the global scale. In two-signal analyses, the ANT scaling factors are also closer to unity in the 5100 and 5605 periods. Continental-scale detection results also demonstrate that ASI, where AO and PDO impacts are strong, has scaling factors of ALL and ANT that are closer to unity particularly in 5100 and 5605 periods (Fig. 11).

As a whole, our updated analysis supports previous findings of Min et al. (2013)—robust detection of ANT signals in temperature extremes on global to continental scales and separation of the ANT signal from the NAT signal preferentially over the northern mid-latitudes. It should be noted, however, that attribution of regional-scale changes in temperature extremes remains challenging. Our examination of the influence of climate variability on observed changes

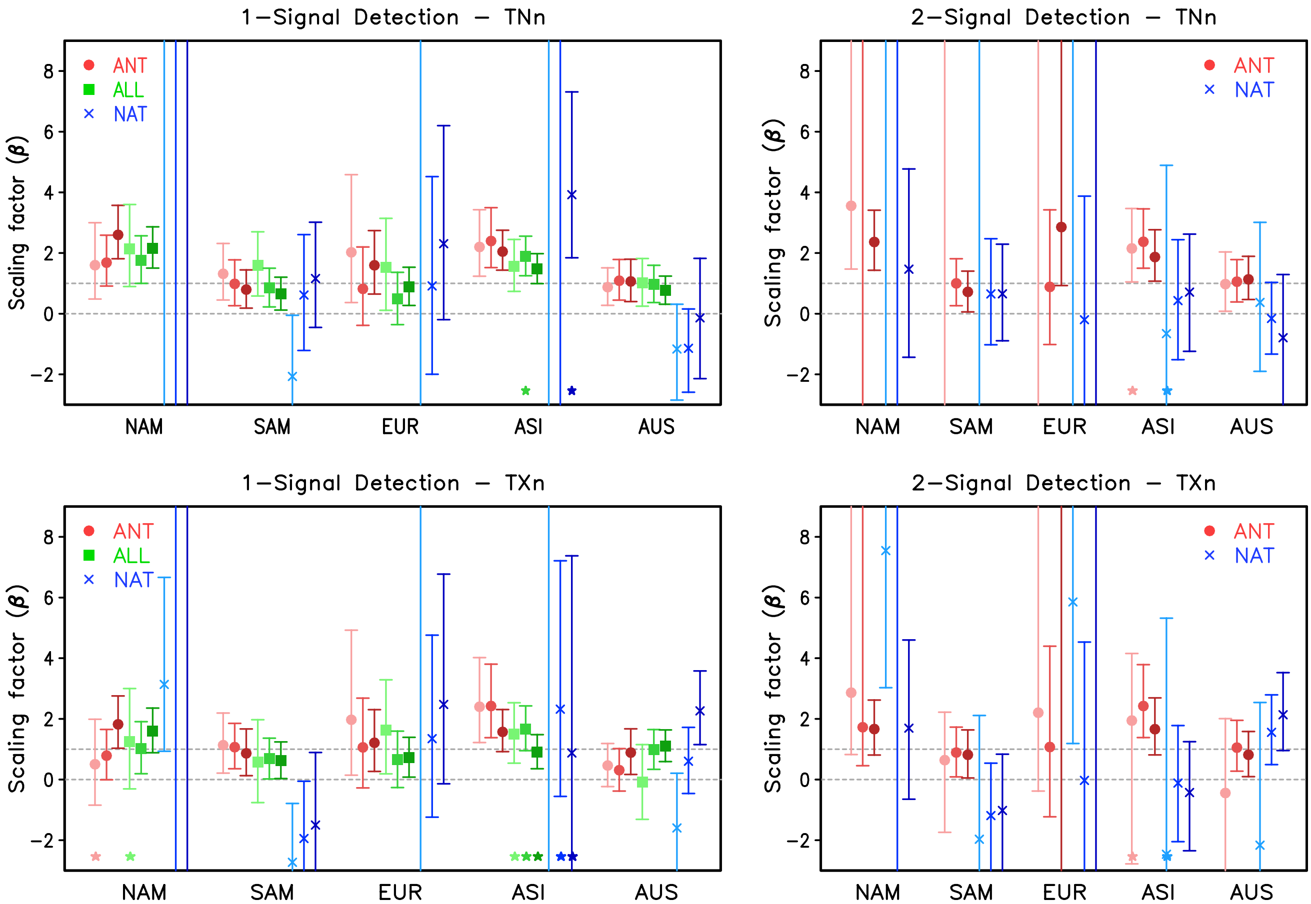

Fig. 11 As in Fig. 7, but when using AO and PDO residual observations

was undertaken, in part, to obtain a better understanding of the differences in attribution results for different continents. However, other local factors such as land use change (Christidis et al. 2013) and urbanization (Ren and Zhou 2014; Sun et al. 2014) that may have affected long-term changes in temperature extremes in some regions should also be considered.

4 Summary and conclusion

This study provides an updated detection and attribution analysis of extreme temperature changes by comparing four extreme indices (TNn, TXn, TNx, and TXx) from HadEX2 observations and ALL and NAT simulations from the CMIP5 multi-model ensemble for three 50-year periods during 1951–2010. Overall results for 5100 are found to be similar to previous findings that were based on older datasets of HadEX and CMIP3. It is found that detection results are not sensitive to either the use of different datasets (HadEX-CMIP3 vs. HadEX2-CMIP5 combinations) or different observed data coverage (HadEX and HadEX2). One exception is that CMIP5 models show better agreement with observed changes in warm extremes (TNx and TXx). Reasons for the improvement in the representation of warm extremes in CMIP5 need to be further investigated.

Our results show that the long-term behaviors of large-area mean extreme temperatures and corresponding detection results are somewhat sensitive to the choice of analysis period (5100, 5605, or 6110). Results based on observed global-mean changes show better agreements between observations and models in the 6110 period, especially for cold extremes. Global results also show that the response to ANT forcing is detectable and clearly separable from the NAT signals for all indices over all periods; also NAT is detected for warm extremes in the 5100 and 5605 periods. Continental-scale analyses show detection of ALL and ANT signals in observed changes in cold extremes over most continents, with better agreement with observations in 6110. ALL and ANT are robustly detected for warm extremes, irrespective of analysis periods.

We also assess the influence of climate variability on detection results, which is found to be important especially for cold extremes. Detection results from using observations with AO and PDO influences removed provide consistent scaling factors across periods, indicating that model-observation discrepancy is partly due to impact of climate variability on observations, which was stronger over Northern mid-latitudes in the winter season. As a whole, our updated results provide strong evidence for human influence on both hot and cold extremes at global and continental scales.

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